

James Evans, Xinning Ma, & Yanshun Zhang
jamwevan@umich.edu, xinningm@umich.edu, & yanshun@umich.edu
University of Michigan
August 10, 2026

Editorial Board
Rose-Hulman Undergraduate Mathematics Journal

Dear Editors,

We are writing to submit our paper, titled *Failure of Converse Theorems of Gauss Sums Modulo ℓ* , for consideration for publication in the **Rose-Hulman Undergraduate Mathematics Journal**. This research was conducted at the University of Michigan between January 8, 2025, and May 8, 2025, as part of our undergraduate studies.

The purpose of this paper is to investigate the failure of the converse theorem for Gauss sums modulo a prime ℓ . We develop a SageMath-based computational framework to identify cases in which distinct multiplicative characters produce identical twisted Gauss sums. Our work confirms previously known counterexamples and discovers new ones, including a counterexample at $(\ell, q) = (2, 49)$ that disproves the conjecture that failures occur only when $q = 2\ell^i + 1$ for some $i > 0$. We expect this work to contribute a clearer understanding of when the converse theorem fails in the modular setting and to show that these failures are more widespread and structurally intricate than previously suggested.

Our sponsor for this submission is:

Name: William Garland
Affiliation: Tutor
Email: william@wjgtutor.com

Please do not hesitate to contact us if any additional information or documentation is needed regarding the paper or its contents. We appreciate your time and consideration, and we look forward to the opportunity to share our work with the broader mathematical community through your journal.

Sincerely,

James Evans, Xinning Ma, & Yanshun Zhang